Jayanth Kumar Dadi

Data Engineer

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PROFESSIONAL SUMMARY:

- **Data Engineer** with **about 7 years** of experience designing, building, and maintaining end-to-end data pipelines, scalable **ETL/ELT** processes, and analytics infrastructure across Finance, Healthcare, Telecom industries using **Azure, AWS**, and **GCP** ecosystems.
- Strong knowledge on **Data Analytics, Text Mining, Apache Airflow, ML Flow, Microsoft Fabric, Machine Learning (ML), Predictive Modelling,** and **Natural Language Processing (NLP).**
- Strong background in **MLOps**, leveraging tools like MLflow, DVC, and Weights & Biases for model experiment tracking and version control.
- Expertise in managing full life cycle of Data Science project includes transforming business requirements into **Data Collection, Data Cleaning, Data Preparation, Data Validation, Data Mining** and **Data Visualization** from structured and unstructured Data Sources.
- Experience working in **Azure Cloud, Azure DevOps, Azure Data Factory, Azure Data Lake Storage, Azure Synapse Analytics, Microsoft Fabric, Azure Analytical services, Azure Cosmos NO SQL DB, Azure HD Insight Big Data Technologies (Hadoop and Apache Spark)** and **Data bricks.**
- Experienced in python data manipulation for loading and extraction as well as with **python** libraries such as **NumPy**, **SciPy** and Pandas for data analysis and numerical computations.
- Experience in **Text Analytics, generating data visualizations** using **R, Python** and creating dashboards using tools like **Tableau.**
- Hands-on with **Snowflake** data engineering (ELT, dimensional modeling, performance tuning) and ML pipeline operationalization; strong Python/Spark/Airflow + MLOps background.
- Expertise in leveraging the **Exploratory Data Analysis** (**EDA**) with all numerical computations and by plotting all kinds of relevant visualizations to do feature engineering and to get feature importance.
- Hands on with creating Reports, Dashboard, aggregates, Hierarchies, Charts, Filters, Quick filters, Table calculation, trend lines calculated measures in Tableau.
- Good experience of setting up Kubernetes clusters on **AWS EKS** and **Azure AKS** for hosting **Apache Airflow**.
- Experience with Data Analytics, Data Reporting, Ad-hoc Reporting, Graphs, Scales, PivotTables and OLAP reporting.
- Proven experience working on Supervised, Unsupervised techniques such as Regression, Classification, Clustering, and Machine Learning (ML) to manage data driven programs.
- Experience and knowledge in provisioning virtual clusters under AWS cloud which includes services like EC2, S3, and EMR.
- Proficient in deploying AI models on cloud platforms, especially **AWS** (**SageMaker**, **Lambda**, **API Gateway**, **CloudWatch**, **S3**) for low-latency, scalable model inference and monitoring.
- Understanding of Data Warehousing principles like **Fact Tables, Dimensional Tables, Dimensional Data modelling - Star Schema** and **Snow Flake Schema.**
- Strong experience in building **fraud detection** and **anomaly detection systems**, leveraging **generative modeling**, **unsupervised learning**, and streaming data frameworks like **Apache Kafka** and **Apache Flink**.
- Experienced in containerizing the enterprise applications using **Docker**.
- Proficient in Data Analysis with sound knowledge in extraction of data from various database sources like **MySQL, MSSQL, Oracle, Teradata** and other database systems.
- Experience in using **GIT** Version Control System.
- Continually updated with the latest **AI trends**, open-source innovations, and research publications to apply cutting-edge advancements in practical use cases.
- Good analytical, problem solving, communication and interpersonal skills, with ability to interact with individuals at all.

TECHNICAL SKILLS:

Programming & Scripting Languages: Python, PySpark, SQL, PL/SQL, Scala, Java, Shell Scripting
Cloud Platforms:

- Microsoft Azure: Microsoft Fabric, Azure Data Factory (ADF), Azure Synapse Analytics, Azure Databricks, Blob Storage, Azure Machine Learning Studio, Azure Data Lake, Delta Tables, Lakehouse
- Google Cloud Platform (GCP): BigQuery, Dataflow, Vertex AI, Pub/Sub, Cloud Composer
- Amazon Web Services (AWS): Amazon S3, Amazon Redshift, AWS Glue, AWS Lambda, Amazon EMR, Amazon SageMaker

AI / Machine Learning: MLflow, DVC, Azure ML Studio, Vertex AI, SageMaker, Scikit-learn, TensorFlow, PyTorch, XGBoost, MLlib, Pandas Profiling, SHAP, Prophet, Optuna, Keras, Hugging Face Transformers, NLTK, OpenCV, Hyperopt, Gensim
Data Engineering & Orchestration: Apache Airflow, Cloud Composer, Azure Data Factory, Databricks Delta Tables, Lakehouse
Databases & Data Stores: Snowflake, SQL Server, Oracle, MySQL, PostgreSQL, MongoDB, Cassandra
BI & Data Visualization: Power BI, Tableau, Looker, Domo, Seaborn, Matplotlib, Plotly
DevOps & CI/CD: Jenkins, Azure DevOps, Terraform, GitHub, Bitbucket, GitLab CI/CD

Operating Systems: Linux (Ubuntu, RHEL), Unix, Windows 10/11

PROFESSIONAL EXPERIENCE:

Client: Healthfirst, New York, NY **Feb 2025 – Present**
Project: HIPAA-Compliant Healthcare Data Platform Modernization and Advanced Analytics Enablement
The project focused on modernizing Healthfirst's healthcare data ecosystem by migrating legacy data integration and reporting pipelines to a cloud-native, scalable architecture using Microsoft Azure and Microsoft Fabric. The initiative enabled secure ingestion, processing, and analytics of clinical, claims, and member data containing PHI, while also introducing advanced machine learning and AI capabilities to support patient risk stratification, utilization analysis, and outcome prediction.
Role: Data Engineer
Responsibilities:

- Designed and orchestrated scalable healthcare data workflows using **Apache Airflow**, ensuring secure, auditable processing of **PHI** in compliance with **HIPAA** and internal governance policies.
- Built automated **ETL pipelines** using **Python**, **PySpark**, and **Airflow** to ingest **clinical, claims, and member data** containing **PHI** into **Azure Data Storage** platforms with enforced access controls.
- Architected and implemented end-to-end data pipelines in **Microsoft Fabric**, enabling ingestion, transformation, and curation of regulated healthcare data for **clinical analytics, reporting, and BI**.
- Migrated regulatory and compliance reporting pipelines from legacy **Azure Data Factory (ADF)** to **Microsoft Fabric**, improving performance while maintaining **HIPAA-compliant auditability**.
- Developed secure data ingestion and migration workflows using **Azure Data Factory**, **Databricks, T-SQL**, **Spark SQL**, **U-SQL**, **SSIS**, and **PowerShell**, ensuring encryption **at rest and in transit** for sensitive healthcare data.
- Designed, developed, and deployed **machine learning models** using **Python**, **TensorFlow**, and **PyTorch** to support healthcare use cases such as **patient risk stratification**, **utilization analysis**, and **outcome prediction**.
- Operationalized LLM/NLP enrichment inside Snowflake using **Cortex.**
- Implemented **AI/ML algorithms** including **regression**, **classification**, **clustering**, and **reinforcement learning** using **Scikit-learn** and deep learning frameworks within regulated healthcare environments.
- Applied **LLM/NLP pipelines** (Transformers + rules + validation) to **normalize messy text fields**, extract entities, and improve downstream matching/analytics quality.
- Built advanced predictive and analytical models leveraging **statistical analysis**, **machine learning**, and **data visualization** to improve **patient outcomes**, **cost containment**, and **care delivery efficiency**.
- Implemented feature engineering and transformations using **Snowpark (Python)** to execute logic in Snowflake.
- Tuned and evaluated models using **Logistic Regression**, **Decision Trees**, **Random Forest**, **Neural Networks**, and Bayesian techniques, ensuring reproducibility and explainability for **clinical and regulatory review**.
- Built ETL pipelines landing curated datasets in **Snowflake**, implementing **Snowflake schema** for analytics and downstream ML.
- Engineered end-to-end **MLOps pipelines** using **Docker**, **Kubernetes**, **MLflow**, **DVC**, **Weights & Biases**, and **Apache Airflow**, enabling compliant model training, deployment, monitoring, and drift detection.
- Deployed machine learning models using **Databricks MLflow,** integrated with feature engineering pipelines built in **PySpark,** supporting predictive analytics at scale.
- Developed applications using **PySpark** to integrate data coming from other sources like ftp, csv files processed using **Azure Databricks** and written into **Snowflake.**
- Built **Streamlit** analytics app for data quality checks / model monitoring dashboards, enabling self-serve exploration for stakeholders
- Orchestrated ML pipelines using **MLflow**, sourcing curated training datasets from **Snowflake** and ensuring lineage, reproducibility.
- Automated **CI/CD workflows** using **GitHub Actions**, **Jenkins**, and **GitLab CI/CD**, supporting versioned, auditable deployments aligned with **healthcare compliance requirements**.
- Containerized and deployed ML models using **Docker** and **Kubernetes** for secure, scalable inference while restricting access to **PHI**.
- Developed **NLP-based classification models** using **LSTM**, **GRU**, and supervised learning to analyze clinical text while applying safeguards for **sensitive patient information**.
- Applied **data quality validation** techniques to monitor **Critical Data Elements (CDEs)**, supporting regulatory reporting accuracy and healthcare compliance standards.
- Designed and maintained **SQL schemas** with **referential integrity** to support compliant healthcare analytics and reporting.
- Programmed analytical utilities using **Python**, **NumPy**, **Pandas**, and **SciPy** for large-scale healthcare data processing.
- Worked in an **Agile environment**, collaborating with clinical, compliance, and security teams to ensure continuous adherence to **HIPAA and PHI handling standards**.
- Supported hybrid cloud healthcare architectures using **AWS (MySQL)** for non-PHI workloads, following enterprise security and compliance guidelines.

Environment: Microsoft Azure (Microsoft Fabric, Azure Data Factory, Azure Databricks, Azure Data Storage), Apache Airflow, Python, PySpark, Spark SQL, T-SQL, U-SQL, SSIS, TensorFlow, PyTorch, Scikit-learn, Docker, Kubernetes, MLflow, DVC, Weights

& Biases, GitHub Actions, Jenkins, GitLab CI/CD, SQL, NLP (LSTM, GRU), Git, Agile, Snowflake, HIPAA/PHI-Compliant Healthcare Data Platforms.

Client: Combined Insurance, Chicago, IL **Jan 2024 – Jan 2025**
Project: Insurance Risk Scoring & Pricing Analytics Modernization on Google Cloud (GCP)
This project modernized Combined Insurance's data and machine learning ecosystem by building cloud-native ML pipelines on Google Cloud Platform (GCP) to support underwriting and claims analytics. The initiative enabled large-scale ingestion and processing of policy, claims, and customer datasets, supporting both batch and near-real-time workflows. The solution introduced standardized MLOps practices (experiment tracking, versioning, CI/CD-style deployments) and automated retraining pipelines to improve operational efficiency, reduce deployment risk, and enable consistent model governance.
Role: Data Engineer
Responsibilities:

- Designed and deployed end-to-end machine learning models using **Python**, **TensorFlow**, and **PyTorch** on **Google Cloud Platform (GCP)** to support insurance **risk scoring, customer analytics, and pricing** use cases with **99.5% uptime**.
- Built scalable training and inference pipelines using **Compute Engine**, **Vertex AI**, and **Google Kubernetes Engine**, reducing model deployment time by **25–30%** across underwriting and claims analytics workflows.
- Implemented experiment tracking and model versioning using **MLflow** and **DVC**, improving reproducibility and reducing rollback issues by **40%** in regulated insurance environments.
- Processed **2–5 TB/day** of policy, claims, and customer data using **PySpark**, **Spark SQL**, **Dataproc**, and **Dataflow**, enabling analytics latency under **2–3 minutes**.
- Automated ML retraining and deployment workflows using **Cloud Workflows**, **Cloud Functions**, and **Cloud Composer**, reducing manual effort by **50–55%** for pricing and fraud models.
- Optimized models for serverless inference using **Cloud Functions**, improving response latency by **20–25%** and lowering operational costs by **15–20%** for real-time quote and recommendation systems.
- Delivered **Streamlit in Snowflake** app for governed, in-warehouse analytics and KPI exploration with role-based access
- Deployed and managed real-time inference using **Vertex AI multi-model endpoints**, reliably handling **20K–50K predictions/day** for customer segmentation and propensity scoring.
- Containerized ML and LLM pipelines using **Docker** and deployed on **Google Kubernetes Engine**, doubling throughput during peak periods such as policy renewals and catastrophe events.
- Built predictive and segmentation models using **K-Means**, **Gaussian Mixture Models**, and **classification algorithms** to improve customer targeting, retention, and cross-sell effectiveness by **8–15%.**
- Designed dimensional data models using **Snowflake schema** in **BigQuery** to support actuarial, claims, and marketing analytics, improving query performance by **25–30%** and reducing data inconsistencies by **30%.**

Environment: Google Cloud Platform (Cloud Storage, Dataflow, Dataproc, Cloud Scheduler), Apache Spark, Python, Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, NLTK, SQL, Snowflake Schema, Git, Agile.

Client: Cross River Bank, Teaneck, NJ **May 2022 – Dec 2023**
Project: Enterprise Data Lake & Analytics Platform on Google Cloud (GCP)
The project focused on architecting and building an Enterprise Data Lake on Google Cloud to centralize and process high-volume data across multiple domains such as lending, payments, partner/fintech integrations, and operational reporting. The solution enabled ingestion of rapidly changing datasets into Google Cloud Storage, transformation using Cloud Dataflow and Dataproc (Spark), and delivery of analytics-ready datasets for reporting and downstream machine learning initiatives.
Role: Data Engineer
Responsibilities:

- Architected and implemented an **Enterprise Data Lake** using **Google Cloud Storage**, supporting ingestion, processing, analytics, and reporting for high-volume, rapidly changing datasets.
- Designed and maintained scalable **ETL pipelines** to ingest, cleanse, and standardize customer data for **KYC and AML compliance**.
- Created and maintained scalable data pipeline architectures using **Cloud Dataflow** and **Dataproc**.
- Developed and optimized complex analytics and transformation jobs using **Dataproc (Spark)**.
- Developed **near real-time data pipelines** to support AML transaction monitoring and suspicious activity detection.
- Implemented incremental and CDC-based ingestion to keep KYC profiles and AML data up to date.
- Built historical data stores to enable back-testing of AML rules and regulatory investigations.
- Enabled cross-border **AML analysis** by handling multi-currency and regional regulatory datasets.
- Performed advanced data analysis and machine learning using **Python**, **Pandas**, **NumPy**, **SciPy**, **Matplotlib**, **Seaborn**, **Scikit-learn**, and **NLTK**.
- Built and evaluated machine learning models including **Linear Regression**, **Naive Bayes**, **Decision Trees**, **Random Forest**, **KNN**, **SVM**, **XGBoost**, **Hierarchical Clustering**, and **DBSCAN**.
- Conducted **statistical and quantitative analysis** on structured and unstructured data to support healthcare and claims analytics.
- Implemented **data cleaning**, **feature engineering**, **feature selection**, and **feature importance** techniques using **Scikit-learn** to improve model accuracy and stability.

- Built distributed **TensorFlow** environments across **CPU** and **GPU** nodes to enable parallel model training.
- Optimized SQL queries and distributed processing jobs, reducing **KYC/AML** data processing time.
- Automated periodic **regulatory reporting datasets** for AML and KYC reviews
- Applied **normalization and denormalization** techniques to optimize performance in **relational and dimensional databases**.
- Designed **logical and physical data models** using **Snowflake schema** for **data warehouses** and **data marts**.
- Created **Entity-Relationship Diagrams (ERD)** to support scalable database design.
- Version-controlled the data science ecosystem using **Git**, enabling collaboration and traceability across teams.
- Automated end-to-end workflows for **model execution, analytics, and reporting**, scheduling recurring jobs using **Cloud Scheduler** and workflow orchestration tools.
- Worked in an **Agile environment**, supporting iterative development, testing, and continuous integration of new requirements.

Environment: Google Cloud Platform (GCS, Dataflow, Dataproc, Cloud Scheduler), Python (Pandas, NumPy, SciPy, scikit-learn, NLTK), Spark, TensorFlow (CPU/GPU), SQL, Git, Agile CI/CD.

Company: Youngsoft India, Hyderabad, India **Jan 2020 – Aug 2021**
Project: AWS Banking Data Lake & Real-Time Analytics Modernization
This project delivered a secure and scalable **enterprise data lake on AWS** to ingest, process, and curate large volumes of banking transactional and customer datasets. The solution enabled both **batch and near-real-time processing**, powering analytics and reporting with high availability and improved performance. The architecture leveraged **AWS S3 + AWS Glue + EMR/Spark** for processing, **Kinesis** for streaming ingestion, **Redshift** for dimensional warehousing, and **Step Functions** for resilient orchestration. Strong governance controls were enforced via **IAM, encryption, and CloudTrail** to meet banking compliance expectations.
Role: Data Engineer
Responsibilities:

- Designed scalable **data ingestion pipelines** using **AWS S3**, **AWS Glue**, and **AWS Lambda**, processing **3–6 TB/day** of transactional and customer banking data.
- Built batch and near real-time processing workflows with **Apache Spark**, **AWS Glue ETL**, and **Amazon EMR**, reducing end-to-end data latency to **under 10 minutes**.
- Architected a secure **enterprise data lake** on **Amazon S3**, supporting analytics and reporting with **99.9% data availability**.
- Orchestrated end-to-end workflows using **AWS Step Functions**, reducing pipeline failures by **35–40%.**
- Modeled financial and risk datasets in **Amazon Redshift** using **star and snowflake schemas**, improving query performance by **30–35%.**
- Implemented **data quality checks** and reconciliation processes, reducing reporting discrepancies by **25–30%.**
- Optimized storage and query performance through **partitioning**, **compression**, and **Redshift tuning**, achieving **20–25% cost savings**.
- Integrated batch and streaming data sources using **AWS Kinesis**, enabling near real-time ingestion and analytics.
- Enforced **data security and governance** using **AWS IAM**, **encryption**, and **AWS CloudTrail**, ensuring compliance with banking regulatory standards.

Environment: AWS (S3, Glue, Lambda, EMR, Redshift, Kinesis, Step Functions, IAM, CloudTrail), Apache Spark, SQL, Python, Data Lake Architecture, Star & Snowflake Schema, Encryption, Banking Data Governance.

Company: mroads, Hyderabad, India **Aug 2018 – Dec 2019**
Project: Customer Analytics, Data Integration (SSIS) & BI Reporting Enablement
The project focused on building a foundational analytics system to support customer behavior analysis and business reporting. The solution involved creating ETL pipelines using SSIS to integrate data across platforms into a centralized database, performing data profiling and validation using SQL, and developing analytics workflows using Python. BI dashboards were developed and published to Tableau Server, with security controls applied to support sector-level and market-level reporting.
Role: Data Analyst
Responsibilities:

- Designed SSIS package to perform extract, transform and load (ETL) data across different platforms and validate the data and achieve the data files into the database.
- Used **Pandas, OpenCV, NumPy, Seaborn, TensorFlow, Kera's, Matplotlib, Sci-kit-learn, NLTK** in Python for developing data pipelines and various machine learning algorithms.
- Configured EC2 instances and configured IAM users and roles and created S3 data pipe using **Boto API** to load data from internal data sources.
- Applied tableau security and published report to **Tableau Server based on Sector Level** & **Market Level.**
- Participated in feature engineering such as feature intersection generating, feature normalize and label encoding with Scikit-learn pre-processing.
- Designed and provisioned the platform architecture to execute machine learning use cases under Cloud infrastructure, **AWS**, **EMR**, and **S3**.

- Worked on outliers identification with box-plot, studentized, K-means clustering using **Pandas, NumPy.**
- Worked on different data formats such as **JSON, XML.**
- Performed **Data Profiling** to assess data quality using **SQL** through complex internal database.
- Isolating customer behavioural patterns by analysing millions of customer data records over a period of time and correlating multiple customers' attributes.
- Used Agile approaches, including Extreme Programming, Test-Driven Development, and **Agile Scrum.**

Environment: Python, SQL, Pandas, NumPy, OpenCV, NLTK, Matplotlib, Seaborn, AWS (EC2, S3, IAM, EMR, Boto3), SSIS, Tableau, JSON, XML, Agile (Scrum)

EDUACTION

Master of Science in Computer Science - Rowan University, Glassboro, New Jersey, USA Aug 2021 - Dec 2022
Bachelor of Technology in Computer Science - Koneru Lakshmaiah University, Andhra Pradesh, India Aug 2015 – May 2019